Revision Note

Paper: Receiver-Domain Behavioral Probing for Backdoor-Resilient Federated GPS Spoofing Detection
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This was a rebuild around one story, not a compression. Nothing was shrunk to fit; sections were removed, merged or rewritten based on whether they advance the argument a reviewer has to be convinced by.

Result: 15 pages, 12 tables/figures, 25 subsections becomes ~6 pages, 1 table and 4 figures, 12 references. The title changed to lead with the mechanism rather than the property, since the mechanism is the contribution.

The story the paper now tells

Self-reported performance is an exploitable attack lever in federated UAV spoofing detection. We propose receiver-domain behavioral probing, which lets the coordinator judge client models on evidence it generates itself. Four result elements carry the whole argument, in order:

attack validity -> competitive comparison -> key differentiator -> limitation and generalization

Everything that did not serve that chain was cut.

Retained, and why

Element	Why it stays
Table I (8-row comparison)	Proves the attack works, inflation makes it worse, median and FLTrust leave residual lift, Multi-Krum is competitive, and ours attributes per client
Fig. 2 (unknown attacker count)	The strongest differentiator: existing rules need f, we do not
Fig. 3 (heterogeneity)	The honest boundary, and the empirical justification for keeping the median backstop
Fig. 4 (trigger generalization)	Supports the scoped claim that the trigger feature need not be known
Fig. 1 (system/threat model)	Redrawn; carries the three attack steps compactly
Algorithm 1	Shortened from 20 lines to 11; the method is the technical center

Removed from the main paper

All of this remains in the repository artifact and is summarized in one or two sentences where it matters.

- **Positioning table against related work** -- replaced by two compact paragraphs, per instruction
- **Layer-ablation table** -- subsumed by Table I, which contains the same conditions
- **Hyperparameter sweep** table and figure -- one sentence in §V-E
- **Trust-degradation table** -- one clause; its decisive case is now Fig. 3
- **Centralized detector-ceiling table** -- two clauses in the Discussion limitations
- **Cost scaling plots** and the two-denominator table -- two sentences in §V-E
- **Adaptive-attacker table and figure** -- three sentences in §V-E
- **Feature-separability table** (10 rows) -- replaced by the phrase "eight satisfy $d >= 0.05$"
- **False-positive breakdown table** -- folded into the 0.3% figure in Table I and §V-A
- **Failed Dirichlet experiment** -- cut entirely; the working partition is described in one sentence
- **Preprocessing table** and per-step row counts -- reduced to two sentences
- **Round-progression, sensitivity and FLTrust-only figures** -- the tables carry the same numbers

Rewritten rather than trimmed

- **Abstract** : 340 -> 192 words. Problem, attack, idea, one attack number, one defense number, one differentiator, one scope sentence. Multi-Krum stays because it frames the differentiator; the adaptive attacker, cost, trigger sweep and false-positive rate are all out.
- **Introduction** : ~1,900 -> 617 words. Related work compressed from four paragraphs of per-paper summary into one gap paragraph and one FLTrust-positioning paragraph. Contributions cut from five bullets to three.
- **System and threat model** : the three attack steps are now labeled A1-A3 and expressed with symbols, with one equation each where it earns one.
- **Results** : reorganised so each subsection maps to exactly one of the four elements.

- **Discussion and Conclusion** : merged into one section.

References

12, down from 12 -- the count was already in range, but two were re-verified against source during the previous revision and the dataset citation was corrected to the Mendeley *Unmanned Aerial System* release (`10.17632/z7dj3yyzt8.3`), which matches the data actually used. The bibliography is now inline (`thebibliography`) rather than BibTeX, so the paper compiles in one pass with no `.bbl` step.

Figures

All four rebuilt at publication quality and re-exported as **vector PDF** rather than PNG:

- single IEEE column width (3.4 in) so none needs a full-width float
- 6.5-8.5 pt type, legible at print size
- consistent colour role across figures (red = attack/undefended, green = proposed, purple = Multi-Krum, blue = FLTrust, grey = median)
- Fig. 1 redrawn from scratch: the previous version had annotations colliding with the uplink arrows and text overflowing its panel. The diagram, threat model and defense now occupy reserved bands that cannot overlap.

One thing we did not do

The paper does **not** claim the proposed method dominates every baseline, because our own Table I would contradict it. Multi-Krum is statistically indistinguishable from behavioral trust alone under the nominal IID setting and costs far less server time. That result is stated plainly in the abstract, in §V-A and in the Conclusion, and is then used to motivate the two properties that do distinguish the method: no attacker-count parameter, and per-client attribution.

Deliverables in this folder

File	What it is
`main.tex`	The paper, IEEEtran `conference` class, inline bibliography
`figures/fig1..4_*.pdf`	The four figures, vector, publication quality
`build_figures.py`	Regenerates all four from the CSVs
`14_conference_results.ipynb`	Executed notebook reproducing Table I and Figs. 2-4
`fl_common.py`, `fl_runner.py`	The shared harness (split, model, attack, 9 aggregation rules, metrics)
`exp_*.py`, `run_all.py`	The experiments behind the reported results
`results/*.csv`	Every reported value, exported

Reproduce with `python run_all.py baselines attackers noniid` then `python build_figures.py` , or set `RUN_EXPERIMENTS = True` in the notebook.